Micro-Project Technical Report

Course: 24MAT382 - Computational Optimization & Applications
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1. Problem Definition & Scenario

Objective: To design an optimal emergency supply distribution network for Campus City Logistics that minimizes operational costs while satisfying all facility demands.

The system must evaluate three potential warehouse locations (North, South, and East) and select exactly two to ensure operational redundancy. These warehouses must distribute supplies to six critical facilities. The model must adhere strictly to a $1,500,000 annual budget, respect individual warehouse capacities, and ensure 100% of daily facility demands are met.

2. Mathematical Formulation

This problem is formulated as a Mixed-Integer Linear Programming (MILP) model.

Sets & Parameters:

- F: Set of critical facilities, W: Set of potential warehouses
- D_i: Annual demand for facility i
- K_j: Annual capacity of warehouse j
- FC_j: Annualized fixed cost for warehouse j
- C_{ij}: Transportation cost per unit from warehouse j to facility i
- B: Maximum annual budget ($1,500,000)

Decision Variables:

- $y_j \in \{0, 1\}$: Binary variable (1 if warehouse j is open, 0 otherwise).
- $x_{ij} \geq 0$: Continuous variable (units shipped from warehouse j to facility i).

Objective Function: Minimize Total Annual Cost (Z):

$$\text{Minimize } Z = \sum_{j \in W} (FC_j \cdot y_j) + \sum_{i \in F} \sum_{j \in W} (C_{ij} \cdot x_{ij})$$

Constraints:

1. **Demand Satisfaction:** Every facility must receive its required demand.
$$\sum_{j \in W} x_{ij} = D_i \quad \forall i \in F$$
2. **Warehouse Capacity:** Shipments from a warehouse cannot exceed its capacity.
$$\sum_{i \in F} x_{ij} \leq K_j \cdot y_j \quad \forall j \in W$$
3. **Redundancy Requirement:** Exactly two warehouses must be selected.
$$\sum_{j \in W} y_j = 2$$
4. **Budget Constraint:** Total costs must not exceed the allocated budget.
$$\sum_{j \in W} (FC_j \cdot y_j) + \sum_{i \in F} \sum_{j \in W} (C_{ij} \cdot x_{ij}) \leq B$$

3. Computational Implementation & Methodology

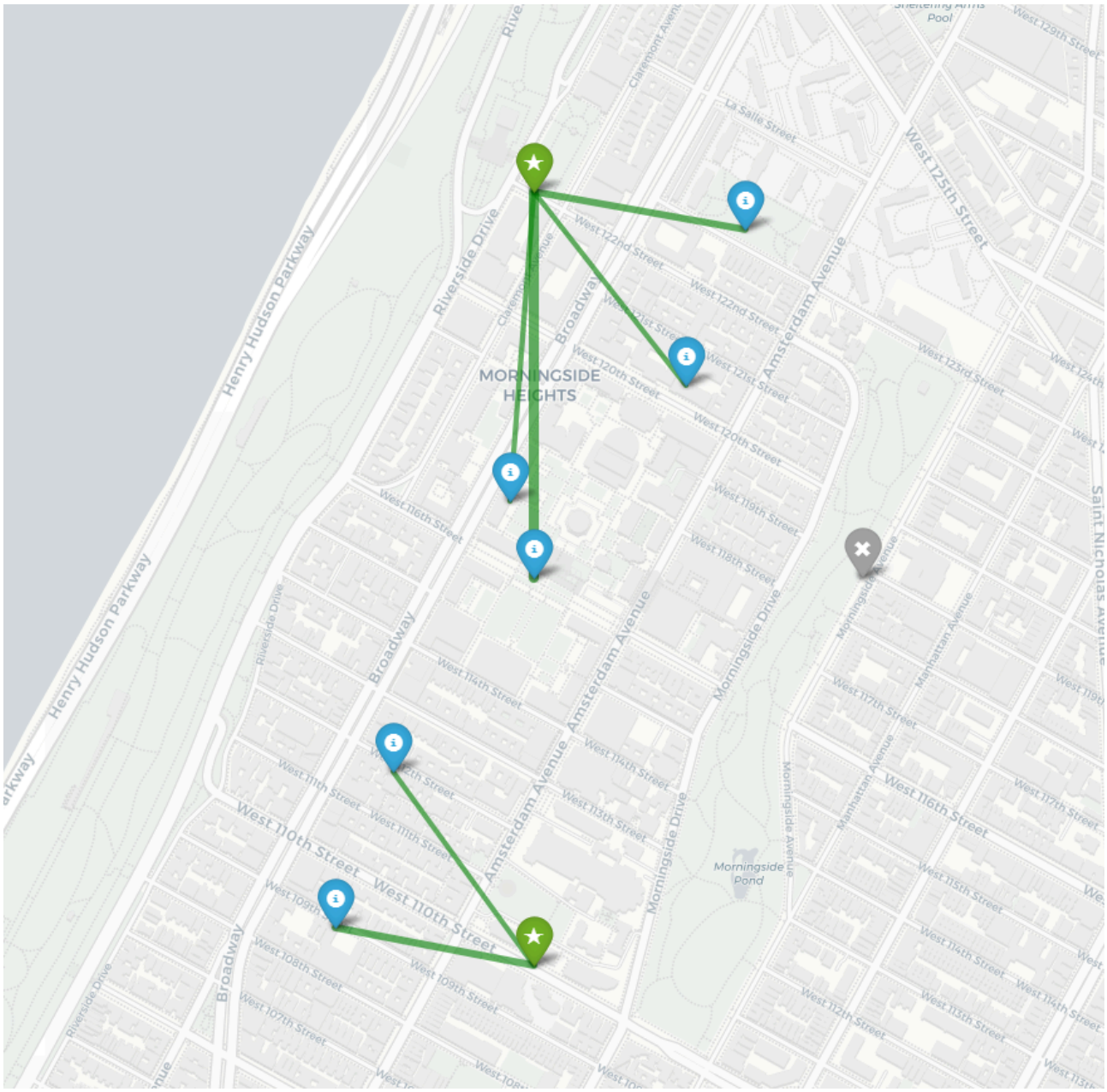
The optimization model was implemented using **Python**.

- **Modeling Logic (`PuLP`):** Utilized to construct the MILP problem, effectively handling both discrete binary variables and continuous variables to guarantee mathematical optimality.
- **Data Processing (`Pandas`):** Used to ingest, merge, and annualize the raw CSV datasets.
- **Visualization (`Folium`):** Used to plot geographic coordinates and dynamically generate an interactive map of optimal supply routes.

4. Results & Constraint Validation

Validation Against Constraints:

1. **Budget Verification:** Total annual cost is **$959,466.05**, generating a surplus of **$540,533.95**. Budget constraint satisfied.
2. **Redundancy Verification:** Exactly two warehouses selected: **North Campus Warehouse** and **South Campus Warehouse**.
3. **Demand & Capacity Verification:** 100% of facility demands met. North warehouse reached 47.5% utilization, South warehouse reached 22.6% utilization.


Henry Hudson Parkway
Riverside Drive
Broadway
MORNINGSIDE HEIGHTS
West 122nd Street
West 121st Street
West 120th Street
West 119th Street
West 118th Street
West 116th Street
West 114th Street
West 113th Street
West 112th Street
West 111th Street
West 110th Street
West 109th Street
West 108th Street
West 107th Street
Amsterdam Avenue
Morningside Drive
Morningside Avenue
Manhattan Avenue
West 125th Street
West 123rd Street
West 117th Street
West 115th Street
West 114th Street
West 113th Street
Morningside Pond
La Salle Street
Saint Nicholas Avenue

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🚀 Initializing Optimization Engine...

✅ OPTIMIZATION SUCCESSFUL
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💰 Total Annual Cost:     $959,466.05
📦 Total Units Delivered: 98,185 units
📈 Avg. Cost Per Unit:    $9.77 / unit
🏦 Remaining Budget:      $540,533.95
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🏢 Warehouse Utilization Report

Warehouse	Status	Fixed Cost	Utilization
WH_NORTH	🟢 OPEN	$322,000	47.5%
WH_SOUTH	🟢 OPEN	$283,500	22.6%
WH_EAST	🔴 CLOSED	$360,500	0.0%

🚚 Optimal Routing Schedule

From	To	Units Shipped	Transport Cost
WH_NORTH	MED_CENTER	29,565	$113,233.95
WH_NORTH	ENG_BUILDING	10,950	$38,653.50
WH_NORTH	DORM_A	17,885	$61,882.10
WH_NORTH	LIBRARY	10,950	$40,186.50